Eric Qin

ericqin0816@gmail.com Katy, TX github.com/ericqin0816

EDUCATION

Seven Lakes High School | Katy, TX | GPA 4.70 | Rank 32 / 885 Expected May 2028

Academic interests: artificial intelligence, software and product engineering, economics and business, and neuroscience.

RESEARCH EXPERIENCE

CruxRNN - Sequence Similarity in Computational Motor Learning

Independent Computational Neuroscience Research | 2026

- Reproduced and extended a 2026 *Scientific Reports* RNN motor-learning study, using a minimal Elman recurrent neural network to test how **sequence similarity** changes the effect of blocked vs. interleaved practice.
- Ran **120 paired experiments / 240 trained RNNs** across Hamming distances 1, 2, 4, and 7; each condition used **15,000 training examples and 750 optimizer updates**, with held-out evaluation and 10,000-resample bootstrap confidence intervals.
- Found the interleaving advantage increased from **+2.1 pp at k=1 to +38.9 pp at k=7**; trend analysis estimated **+6.12 pp per additional differing sequence position** (95% bootstrap CI: 5.38-6.85 pp).

RobustDrive - Failure Detection & Uncertainty in Autonomous Driving

Independent Machine Learning Research | In progress

- Developing an end-to-end **behavioral-cloning system** that trains a CNN from scratch on simulator camera images and steering demonstrations to control a virtual vehicle.
- Building a controlled robustness benchmark across **lighting shifts, blur, noise, shadows, occlusion, and image displacement** to quantify steering degradation under visual distribution shift.
- Implementing **Monte Carlo dropout**, uncertainty-based failure detection, and selective prediction to test whether uncertainty can identify high-error steering outputs and **confidently wrong predictions**.
- Engineering a reproducible ML pipeline with **sequence-aware splitting, side-camera recovery data, augmentation, YAML configs, checkpointing, automated evaluation, pytest, and GitHub Actions CI**.

Residential Solar + Battery Economics Model

Independent Research | In progress

- Developing a quantitative model to estimate the **break-even electricity rate** at which residential solar + battery systems become economically favorable for **Texas homeowners**.
- Testing how system cost, household electricity use, utility rates, battery assumptions, and other inputs shift the threshold across homeowner scenarios; building the study through **literature review**, methodology design, data collection, **sensitivity analysis**, and validation.

TECHNICAL & PRODUCT EXPERIENCE

ClimbIQ - Speed Climbing Video Analysis Platform | Creator & Engineer

- Built a local-first video analysis platform using **React**, **TypeScript**, **MediaPipe Pose**, Canvas, and Web Audio to identify race events, estimate climber motion, and generate reviewable split times.
- Implemented multi-signal event detection combining **timing-light changes**, **audio cues**, and **athlete motion**, with frame-by-frame review and correction for ambiguous results.
- Evaluated the analysis pipeline across **11 source recordings** and **30 controlled variants**, testing compression, resizing, exposure, frame-rate changes, missing audio, and trimming.

Worktree - Collaborative AI Developer Workspace

Co-founder | Marketing & Product

- Lead marketing, product positioning, and user discovery for a desktop workspace where developers and AI coding agents collaborate across shared projects and threads.
- Spoke with **20+ prospective users** to study developer workflows and gather product feedback; connected with a **YC-backed founder** through targeted outreach.

CalcQuest - Adaptive Calculus Learning Game | Character Designer

- Designed **3 original enemy concepts** for an educational RPG focused on AP Calculus AB/BC practice; collaborate with the team on character iteration, game visuals, and additional gameplay assets.

WORK EXPERIENCE

CZT Energy USA | Operations & Quality Control Intern Summer 2026

- Automated **English-Chinese translation**, PDF processing, Excel organization, and internal forms using **AI-assisted workflows**; supported marketing and executive operations.
- Managed **quality-control workflows** at a secondary worksite and trained employees; supported inventory, purchasing, expense, and employee records.

Katy ISD | Water Safety Instructor 7 hours/day | 4 weeks

- Taught children across a wide range of ages and skill levels, adapting technique instruction and feedback to each swimmer.

Private Swim Instruction | Founder & Instructor

- Ran an independent swim-instruction business serving Chinese families; taught **20+ swimmers** across **50+ lessons** while managing scheduling, parent communication, payments, and instruction.

LEADERSHIP & SERVICE

Hearts for Elders | Event Coordinator

- Coordinate volunteer events and outreach across **10+ chapters** and **20+ senior homes**.
- Contacted **20+ organizations** and **20+ companies** to pursue partnerships, sponsorships, and community outreach opportunities.

Seven Lakes High School Student Council | E&E Officer

- Help lead a **600+ member council** and support annual fundraising exceeding **$100,000** for community initiatives, including local shelter meals and supplies.
- Built a **barcode-based attendance system** with Google Sheets and **Google Apps Script** to automate meeting check-in and attendance tracking.

ADDITIONAL PROJECTS

PolyShield - AI-Agent Security Gateway: Co-developed an **MCP security gateway** for AI coding agents with **policy-based tool controls**, approval flows, and auditing; helped build the TypeScript/Next.js dashboard MVP.
Strata - AI Market Intelligence & Backtesting Engine: Built an AI-assisted market research workstation with market signals, strategy simulation, **backtesting**, **portfolio analysis**, and Discord-based tools.

ATHLETICS & ACTIVITIES

Competitive speed climbing: USA Climbing Youth Nationals speed qualifier; personal best 10.51 seconds. Train and use video/timing analysis to evaluate performance.
Swimming: Seven Lakes varsity swimmer; competitive swimmer and certified swim instructor with both public-program and private teaching experience.

TECHNICAL SKILLS & INTERESTS

Programming & web: Python, TypeScript, JavaScript, Java, React, Next.js, Vite, Google Apps Script, GitHub, Vercel.
Computer vision & data: PyTorch, OpenCV, MediaPipe Pose, NumPy, Pandas, scikit-learn, Canvas, Web Audio, Supabase, Excel, Google Sheets.
Languages: English, Mandarin Chinese.
Interests: AI systems, computer vision, software and product engineering, economics and business, neuroscience, and applied research.